

Monitoring rockmass performance systems and geotechnical instrumentation

PURPOSES OF MONITORING ROCK MASS PERFORMANCE

Monitoring is the surveillance of rock structures, either visually or with the help of instruments. Monitoring of rock mass performance in an excavation may be carried out for the following main reasons:

- to maintain safe operational procedure for personnel & Equipment;
- to provide advance notice for instability or hazards;
- to get geotechnical information of rock or soil mass such as strength, deformability, anisotropy etc. for design/determination of supports and remedial technique;
- to analyse failure mechanism of rock or soil mass in which opening is constructed;
- to check change in hydrological behaviour of rock mass;
- identification of parameters causing instability of rock structures;
- identification of the change of stress in rock mass;
- measurement of the response of a rock mass to natural and man-made disturbance;
- to check the validity of the assumptions, conceptual models and values of soil or rock mass properties used in design calculations;
- to verify effectiveness of support and remedial measures.

Stability is a critical safety and production issue for mine structures. Major failure can occur seemingly without any visual warning, causing loss of lives, damage to equipment, and disruption to the mining process. Monitoring systems are employed to predict impending instabilities and failure.

Monitoring to check the rock mass response and, as a consequence, adjust the overall mine design or take remedial measures is equally important. Rock masses are extremely complex media whose engineering properties are difficult, if not impossible, to predetermine accurately ahead of excavation. It will also be clear that the models used to predict the various aspects of rock mass response to different types of mining procedure, are based on idealisations, assumptions and simplifications. It is vitally necessary, therefore, to obtain checks on the accuracy of the predictions made in design calculations. Monitoring the behaviour of the rock mass surrounding the mining excavation is an integral part of a mine rock mechanics programme, and provides the feedback necessary to ascertain proper design. In some cases, the design may be based largely on the results obtained by monitoring trial excavations or the initial behaviour of the prototype excavation, with little or no reliance being placed on pre-excavation design calculations. The use of field observations of the performance of structures is central to the general practice of geotechnical engineering in which it is known as the observational method. Such as, the NATM integrates the principles described as a "design as you monitor" approach, based on observed convergence and divergence in the lining and mapping of prevailing rock conditions.

Monitoring systems used in conjunction with underground mining operations can be ranging from simple to highly sophisticated and expensive. However, it should be remembered that valuable conclusions about rock mass response can often be reached from visual observations and from observations made using very simple monitoring devices.

PLANNING MONITORING OF ROCK MASS

Dunnicliffe and Schmidt have listed the following tasks as essential in systematic rock monitoring:

- Definition of benefits and purposes
- Planning of programme in accordance with rock conditions
- Selection of monitoring parameters
- Selection of instruments
- Writing of specifications for procurement and installation
- Procurement and installation
- Reading, maintenance and protection of instruments
- Data processing and presentation
- Interpretation
- Implementation

SELECTION OF MONITORING PARAMETERS

Instrumentation is done after ascertaining what parameters to measure or detect.

Items that may be monitored in an underground mining operation include

- change in load on support
- change in static stress in rock mass
- measurement of displacement in excavation
- wall- wall and roof-floor convergence
- subsurface rock displacement
- change in hydrological properties in rock mass
- elevation of phreatic surface (gw)
- water pressure
- water flow
- ground vibration & dynamic stress in rock mass
- seismic/micro-seismic/acoustic behaviour of rock mass

GEOTECHNICAL INSTRUMENTATION

A great number of methods are available for rock structural monitoring. These methods can however generally be classified into geodetic (surveying) methods and geotechnical methods. Geodetic methods are mainly used to monitor deformations while geotechnical methods can be used to determine some other important parameters beside deformations. The two types of methods complement each other in most of the times in terms of the types of information that they can obtain. It should be noted here that the above classification of the monitoring methods is

mainly used in the fields of surveying and geodesy. The engineers and geologists often refer all the methods as geotechnical methods.

CHOOSING INSTRUMENTATION

Critical Parameters: Each project presents a unique set of critical parameters. The designer must identify those parameters and then select instruments to measure them. What information is required for the initial design? What information is required for evaluating performance during and after construction? When the parameters are identified, the specification for instruments should include the required range, resolution, and precision of measurements.

Complementary Parameters: The behavior of a rock mass typically involves not one, but many parameters. In some cases, it may be sufficient to monitor only one parameter, but when the problem is more complex, it is useful to measure a number of parameters and to look for correlation between the measurements. Thus it is common practice to choose instruments that provide complementary measurements.

Ground Conditions: Ground conditions sometimes affect the choice of instrument.

Environmental Conditions: Temperature, dust, inflammable gas and humidity also affect instrument choice.

Personnel and Resources at the Site: Consider the personnel and resources at the site when choosing instruments. Do technicians have the skills required to install and read a particular type of instrument? Are adequate support facilities available for maintenance and calibration of the instrument?

Data Acquisition: An automatic data acquisition system may be required when

- There is a need for real-time monitoring and automatic alarms;
- Sensors are located at a remote site or in a location that prevents easy access;
- There are too many sensors for timely manual readings;
- Qualified technicians are not available.

If a data acquisition system is required, the choice of instruments should be narrowed to those that can be connected to the system easily and inexpensively.

Instrument Quality: The difference in cost between a high-quality instrument and a lesser-quality instrument is generally insignificant when compared to the total cost of installing and monitoring an instrument. It is false economy to install a cheaper, less reliable instrument.

Instrument Performance: Instrument performance is specified by range, resolution, accuracy, and precision. The economical designer will specify minimum performance requirements, since the cost of an instrument increases with resolution, accuracy, and precision.

- *Range* is defined by the highest and lowest readings the instrument is expected to produce. The designer typically specifies the highest values required.
- *Resolution* is the smallest change that can be displayed on a readout device. Resolution typically decreases as range increases. Sometimes the term "accuracy" is mistakenly substituted for resolution. Resolution is usually many times better than accuracy and is never expressed as a plus/minus value.
- *Accuracy* is the degree to which readings match an absolute value. Accuracy is expressed as a $\pm$ value, such as $\pm 0.5\text{mm}$, $\pm 1\%$ of reading, or $\pm 1\%$ of full scale.
- *Precision or repeatability* is often more important than accuracy, since what is usually of interest is a change rather than an absolute value. Every time a reading is repeated, the value returned by the instrument is slightly different. Precision is expressed as a $\pm$ value representing how close repeated readings approach a mean reading.

Modes of operation: The modes of operation of the sensing, transmission and read-out systems used in monitoring devices may be mechanical, optical, hydraulic or electrical.

- *Mechanical* systems often provide the simplest, cheapest and most reliable methods of detection, transmission and read-out. Mechanical movement detectors use a steel rod or tape, fixed to the rock at one end, and in contact with a dial gauge or electrical measuring system at the other. The two most frequently used mechanical transducers are dial indicators and micrometers. A dial indicator converts liner movement of a spring-loaded plunger to the movement of a pointer that rotates above a dial. Typical ranges of dial indicators are about 50 mm while long-range dial indicators of up to 300 mm can be obtained. The reading resolutions of dial indicators are generally within $+0.025 - +0.0025$ mm. A micrometer measures displacement by measuring the rotation of a finely threaded plunger when it travels in or out of housing. Fractional revolutions are measured with the assistance of a vernier. Accuracies of micrometers are limited to about $+0.025$ mm. The range of a micrometer can be extended to 150 mm. Micrometers are more robust than dial indicators.

The main disadvantage of mechanical systems is that they do not lend themselves to remote reading or to continuous recording.

- *Optical* systems including electro-optical (EDM) and laser-based systems are used in conventional, precise and photogrammetric surveying methods of establishing excavation profiles, measuring movements of excavation boundaries, and recording natural and mining-induced fractures. These methods are also widely used in monitoring surface subsidence associated with underground extraction. Satellite-based Global Positioning Systems (GPS) may also be used for the latter purpose. In the past, considerable use was made of photoelastic plugs and discs to monitor stress changes in the rock surrounding excavations and in support elements.
- *Hydraulic and pneumatic* diaphragm transducers are used for measuring water pressures, support loads, cable anchor loads, normal components of stress and settlements.

In all cases the method of operation is the same. The quantity measured is a fluid pressure which acts on one side of a flexible diaphragm made of a metal, rubber or plastic. Twin tubes connect the read-out instrument to the other side of the diaphragm. To take

readings, air, nitrogen or hydraulic oil pressure is supplied from the read-out unit through one of the tubes to the diaphragm. When the supply pressure is sufficient to balance the pressure to be measured, the diaphragm acts as a valve and allows flow along the return line to a detector in the read-out unit. The balance pressure is recorded, usually on a standard Bourdon pressure gauge or a digital display.

- *Electrical* devices probably provide the most common basis of the instruments presently used to monitor the performance of rock masses surrounding mining structures, although mechanical systems still find widespread use in displacement monitoring. Electrical systems generally operate on one of three basic principles.
- *Electric resistance strain gauges* operate on a principle discovered by Lord Kelvin, namely that the resistance of a wire changes proportionately with strain. In strain gauge systems, strain gauges made from thin wire or foil, are bonded to a rock, concrete or steel surface. Changes in the strains in the host material are accompanied by changes in the strains, and resistances, of the strain gauges which are read using a Wheatstone bridge circuit. Most stress meters, load cells, many pressure transducers and some types of inclinometer also use electrical resistance strain gauges. The major disadvantages associated with the use of electrical resistance strain gauges for rock mechanics applications are
 - a) it is difficult to obtain and maintain a good bond between the strain gauge and the rock,
 - b) the strains are measured over relatively short gauge lengths, and
 - c) temperature effects cannot always be eliminated.
- *Linearly Varying Displacement Transducers (LVDT)* is also called linear variable differential transformer, consists of a movable magnetic core passing through one primary and two secondary coils (Fig. 1). When an AC voltage, called the excitation voltage, is applied to the primary coil, an AC voltage in each secondary coil is generated, with a magnitude depending on the relative position between the magnetic core and each of the secondary coils. The displacement of the magnetic core can therefore be measured by measuring the changes in the voltages of the secondary coils. The requirement of the AC power supply can be avoided by using DC power together with an amplitude regulator. Most LVDTs used in practice use this approach since it is often troublesome to get AC power supply in the field.

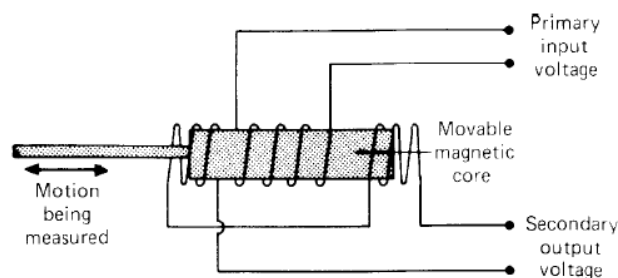


Fig. 1: Schematic of a LVDT

- *Vibrating wire transducers:* A steel wire is clamped at its two ends (Fig. 2). The frequency of vibration of the wire changes with its tension. Small displacements between the ends of the wire can therefore be measured by measuring the frequency changes. The frequency of vibration of a tensioned wire in terms of the wire stress is

$$f = \frac{1}{2L} \sqrt{\frac{\sigma_w}{\rho}}$$

where, f is the natural frequency; L is the length of the vibrating wire; σ_w is the tensile stress in the wire and ρ is the density of the wire material.

The electrical coil usually serves two purposes, plucking the wire and sensing the vibration of the wire in the same time. Some vibrating wire transducers can continuously carry out the measurement while the others measure only at discrete times.

This value may be used to determine the pressure acting on a diaphragm to which one end of the wire is attached, or the axial load on a load cell in which the vibrating wire is mounted. Thus vibrating wire systems are used in piezometers, soil pressure cells, stressmeters and load cells.

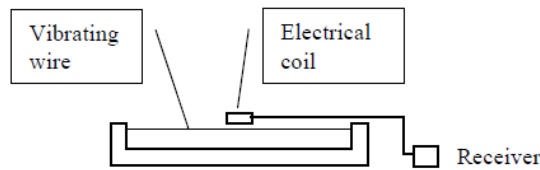


Fig .2: Schematic of vibrating wire transducer

- *Potentiometer:* A potentiometer consists of a fixed resistance strip and a movable slider that makes electrical contact along the resistance strip (Fig. 3). The measured resistance or voltage changes with the position of the contact point.

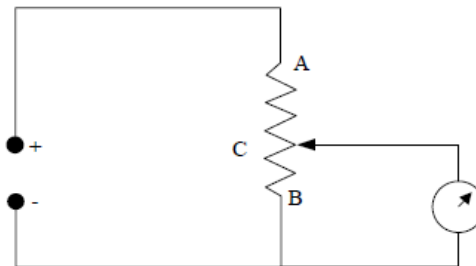


Fig. 3: Schematic of potentiometer

- *Self-inductance* instruments are based on the mutual inductance of a pair of coaxial solenoids forming a circuit of resonant frequency

$$f = \frac{(LC)^{-\frac{1}{2}}}{2\pi}$$

where L is the self-inductance and C the capacitance of the solenoids. A relative displacement, d , of the inner and outer solenoids produces a change in the measured resonant frequency, f , such that

$$d = K \left(1 - \frac{f_0^2}{f^2} \right)^{\frac{1}{2}}$$

where f_0 is the calibrated frequency for the null position and K is a constant. A number of borehole extensometers and inclinometers work based on this principle. These instruments may be read remotely using radio-telemetry.

Survivability: Instruments must be able to survive for duration of their prospective use. This is not a simple requirement, as experience indicates beyond question that field conditions are almost invariably more severe than anticipated.

Cost: It must not be excessive or unreasonable.

MONITORING OF ROCK MASS

The parameters of a structure that need monitored are many but the most common ones are convergence/deformation, load, stress and ground water pressure.

Convergence monitoring on boundary of excavation

Convergence, or the relative displacement of two points on the boundary of an excavation, is probably the most frequently made underground measurement. The measurement is made variously with a telescopic rod, invar bar or tape under constant tension, placed between two measuring points firmly fixed to the rock surface (Figure 4). A dial gauge, micrometer, or an electrical device such as an LVDT, potentiometer or vibrating wire gauge is used to obtain the measurement of relative displacement. When interpreting convergence measurement results, it is important to consider the time rates of convergence and any changes in those rates, as well as the total convergence values.

COMMON GEOTECHNICAL INSTRUMENTS

A list of some geotechnical instruments for typical measurement is given below.

Measuring parameters

1. Roof – Floor & Wall-wall

Displacement/Deformation

Instruments

- Convergence
- Indicator/Recorder
- Tape/Tube Extensometer
- Rod/Wire
- Tell – Tale
- Surveying instrument
(Profilometer, Distomat, Levelling Eqp.)

2. Displacement/Deformation within Rock Mass

- Rod/Wire borehole Extensometer
- Vibrating wire borehole Extensometer
- Potentiometer borehole Extensometer
- Magnetic borehole Extensometer
- Sonic borehole Extensometer
- Inclinator
- Shear strip
- TDR-Time-Domain Reflectometry

3. Borehole Visual Inspecting Device

- Fiberoptic Bore Scope
- Video Bore Scope
- Bore Hole Television Camera

4. Load on Support

- *Load Cell (Strain gauge/Hydraulic/Vibrating
Wire/Potentiometer based)
- * Central Hole Load Cell (for Rock bolt)
- * Instrumented Rock Bolt

4. Stress Change in Rock mass

- Borehole Pressure Cell
- Vibrating Wire Stressmeter (Stress Cell)

5. Ground Water level

- Open pipe Piezometer

6. Ground Water Pressure

- Piezometer (Pneumatic/Hydraulic/VW)

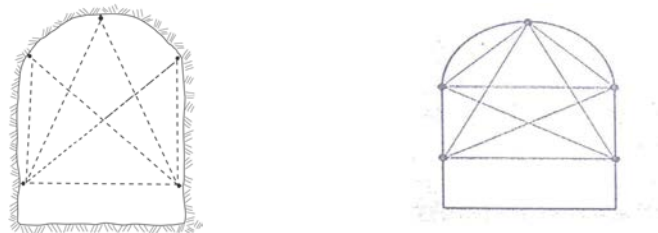


Fig. 4: showing typical convergence array in a tunnel

Mechanical convergence monitoring stations are robust, simple and inexpensive to install, and the results obtained are usually reliable. However, because of potential obstruction to accesses and working places, they are generally not suitable for remote or continuous reading. Regular manual reading of large numbers of stations can be time-consuming.

Convergence monitoring using multiple-point borehole extensometers

Among the most useful measurements of rock mass performance are those made using multiple-point borehole extensometers (MPBXs) (Figure 5). A single-point borehole extensometer gives the relative displacement between an anchor point inside the rock mass and a measuring point, generally located at the excavation boundary. A multiple-point extensometer can also give the relative displacements between several points at different depths in the borehole. In this way, the distribution of displacements in relatively large volumes of rock can be recorded. MPBX measurements are particularly useful in recording bed sag and separation in bedded strata above coal mining roadways. These data are generally more useful than the results of convergence measurements which only give relative surface displacements and may be influenced by surface conditions.

MPBX installations may use a variety of techniques and materials. The extensometer may use stainless steel wires or rods made of aluminium, carbon fibre composites, fibreglass, invar, mild steel or stainless steel. The rods may be anchored with mechanical (e.g. expansion shell), grouted, hydraulic or snap ring anchors. Displacements between the anchor and reference head may be measured mechanically by a micrometer or dial gauge, electronically using potentiometer, LVDT, self inductance coil, strain gauged cantilever or vibrating wire displacement transducers, or by a magnetostrictive (sonic probe) transducer.

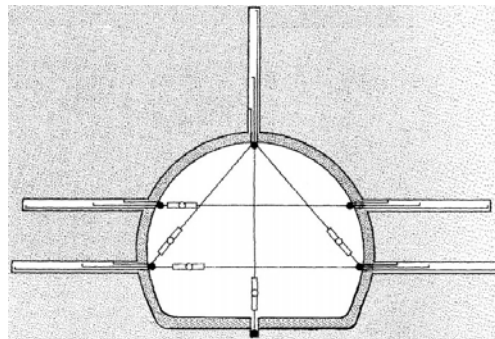


Fig. 5: Multiple-point borehole extensometer (MPBX) installations with convergence measurement between MPBX heads.

Stress change monitoring

The vibrating wire stress meter is used widely for stress monitoring (Figure 6). This instrument was developed originally as a low-cost device for monitoring stress changes in coal, but it has since found use in a variety of hard- and soft-rock applications. From the measurements, the principal stress changes in the plane normal to the instrument axis and their orientations may be determined.

Load change monitoring

The load is most commonly monitored in an element of structural support, i.e. under steel sets in tunnelling, props in mining and anchors in supported excavations. A large variety of mechanical,

hydraulic, vibrating wire and electrical resistance load cells are available which are both rugged and durable. Load cells capable of being read remotely are available based on hydraulics, bonded electrical resistance strain gauges and vibrating wire gauges. Load cells are usually re-usable.

Monitoring of ground water level and pore water pressure

Ground water level can usually be measured through the use of a standpipe. Various instruments that are based on the use of mechanical, electrical, acoustic and pressure sensing gages can be used for this purpose. The commonly used ones include steel tape, electrical dipmeter, audio reader, and piezometers. Piezometers are instruments for the measurement of ground water pressures. They are usually used in an open standpipe, sealed in filled embankment, or driven into ground. The central part of a piezometer is a pressure gage, either a mechanical, electrical, hydraulic or pneumatic transducer. Therefore, we have the vibrating wire piezometers, twin-tube hydraulic piezometers, pneumatic transducers, electrical resistance piezometers, etc.

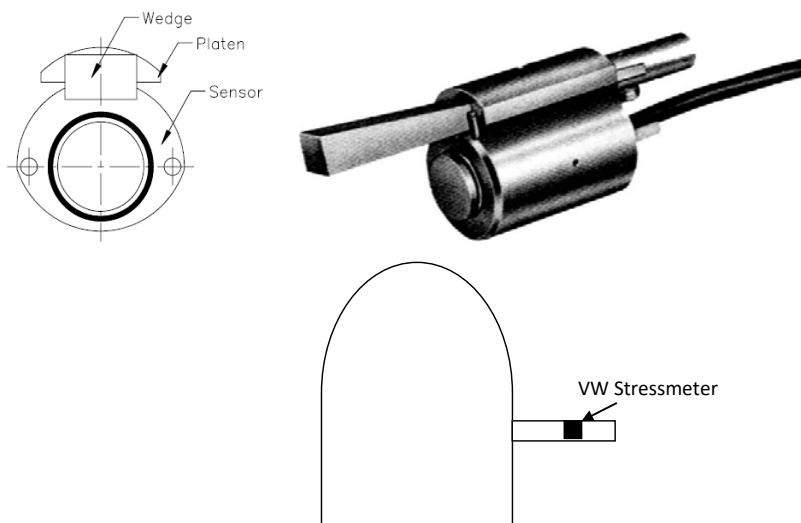


Fig.6: Stress change measurements using vibrating wire stress meter

MONITORING DATA PRESENTATION: GRAPH & MAP/SECTION

Plotting or tabulating of measured data done for obtaining the following information:

- (a) Co-ordinate of Measuring Location/Point
- (b) Displacement
 - Displacement Vector
 - Cumulative Displacement
 - Incremental Rate
- (c) Load – Rate of Change
- (d) Stress – Rate of Change
- (e) Hydrological Prop. – Ground water
 - Water Pressure
 - Flow Rate
- (a) Seismic/Micro-Seismic/Acoustic Activities –
 - Change in Frequency & Magnitude
 - Change in Spectral character
 - Location of Foci

CRITERIA FOR EVALUATION OF MONITORING DATA

In general, displacements measured at several locations in an opening are the best criterion for establishing the stability of an opening for its given size, geology and support system. When considering the magnitude of displacements, both the depth at which the rock mass is loosening and its area of extent must be known to determine the severity of the problem and the remedial measures to be taken. A number of criteria can be used which have been found appropriate for evaluating displacements in underground excavations. A single criterion is not adequate for evaluating the stability of underground workings. Further more, displacements which indicate local instability such as loosening of a thin rock slab, must be distinguished from those which indicate a more widespread and deep-seated condition affecting the stability of the entire excavation.

The following criteria can be used to determine whether displacements are indicative of either stable or potentially unstable behavior:

1. Magnitude of displacement with respect to
 - (a) displacement as predicted from elastic theory;
 - (b) measured displacement in well-supported sections of the opening, in comparable rock conditions.
2. Rate of displacement.
3. Displacement capacity of the support system and of the rock mass.

Based on the monitoring observation data of instability suitable remedial measures are taken to stabilize the underground rock excavations.

***Note:** The contents of this course material are compiled from various rock mechanics books, published papers, unpublished reports and from the internet sources. The contents are only for educational purpose and do not qualify as a publication material elsewhere.*

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